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CAMEF: Causal-Augmented Multi-Modality Financial Forecasting

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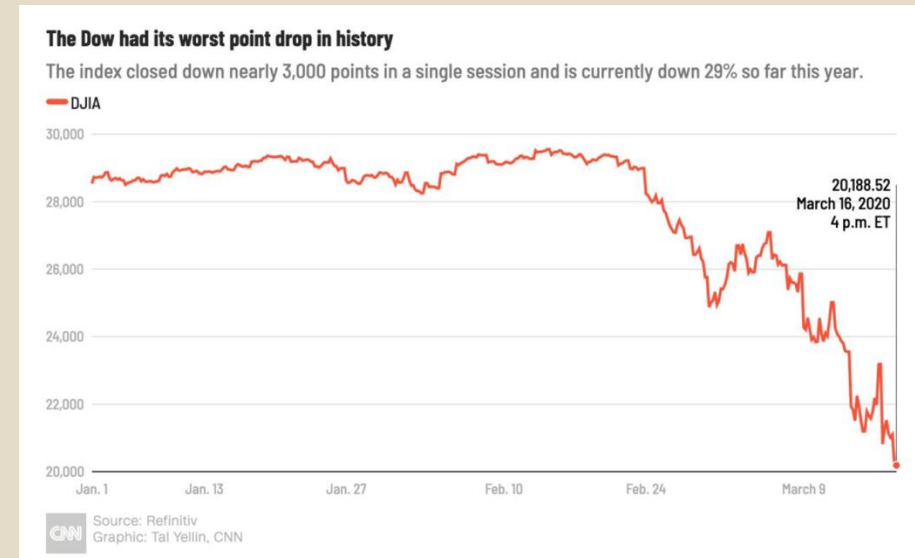
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Motivation

- Major Macro Economic Releases Always Produce Significant Market Impacts
- Case Study: The March 16, 2020 FOMC Shockwave



FOMC: Fed funds rate cut to 0% - 0.25%



The Dow index tumbles nearly 3,000 points
- - worst point drop in history

Motivation

- **Other Key Macro-Economic Releases with Strong Impact [1-8]:**
 - ① FOMC Meeting ② Unemployment Insurance ③ Employment Situation Report
 - ④ CPI Report ⑤ PPI Report ⑥ GDP Advance Report
- **Financial traders worldwide trade on these releases, and thus shaping the upcoming market trend.**



**Key Macro-Economic
Releases**

August 3-7, 2025



**Traders trading the
Information**

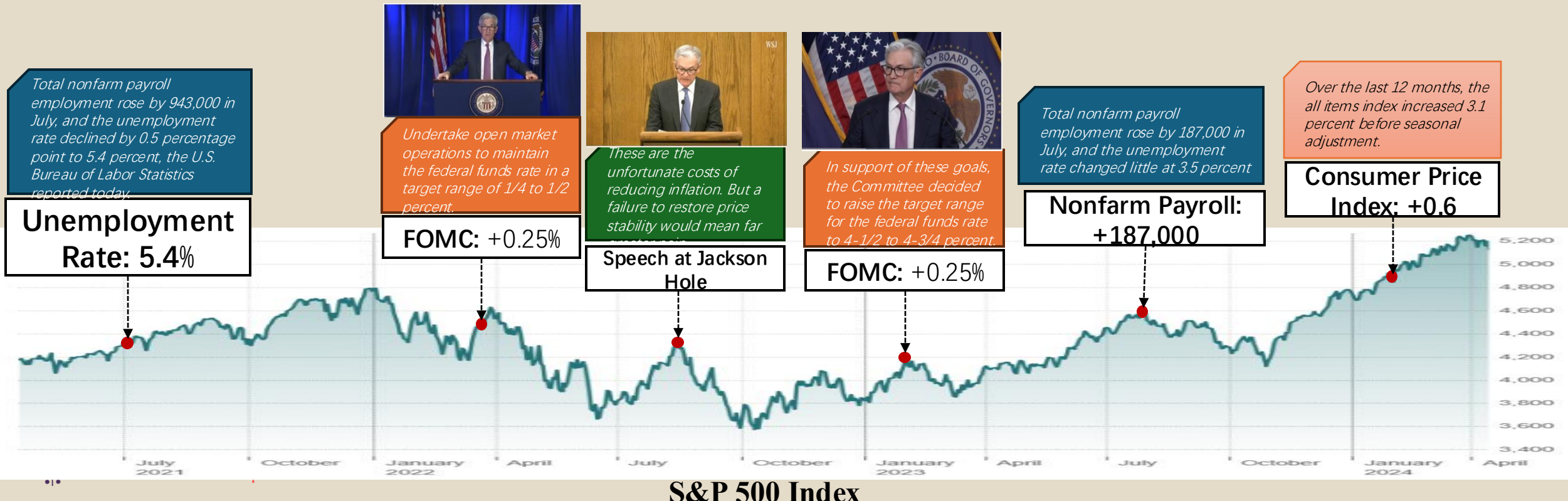


**Thus "Created" the
Market Trend**

The Research Question

Event-Driven Multimodal LLM for Financial and Economical Forecasting

- How does these key EVENTS drive the financial?



Why Study This Topic

- **Forecasting financial trends is crucial for both policy-makers and investors [9].**



Investors

Guide investment decisions for better returns.



Policy Makers

Inform monetary and fiscal policies for stability [10].

Related Work

➤ Prior research on event-driven financial forecasting generally falls into three technical directions:

① Forecasting the market trend based on **time series patterns**

② Forecasting the market trend based on **textual information**

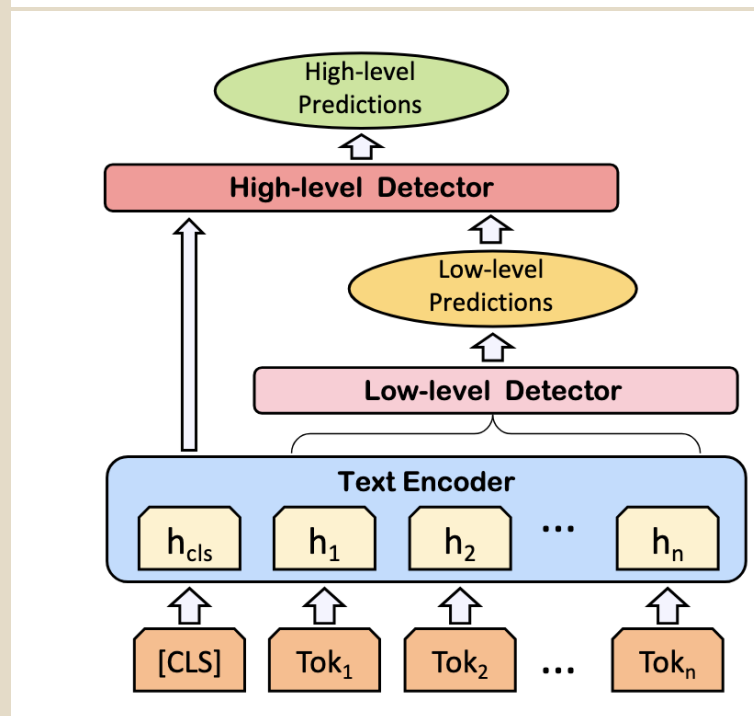
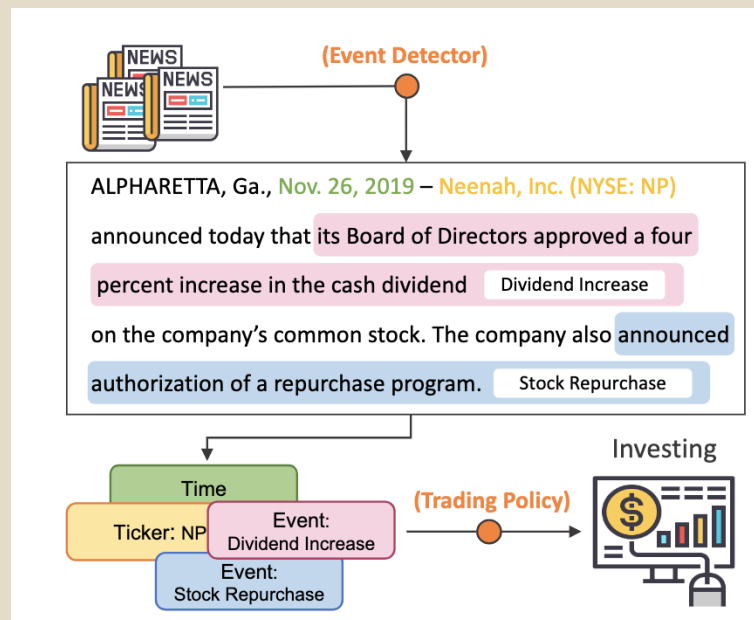
③ Leverage both **time series patterns** and **textual information**

Related Work - Time Series Forecasting

- **Predicts future trends by learning patterns from historical market data.**
 - ① **Traditional Models:** Linear Regression [11], ARIMA [12], GARCH [13].
 - ② **Deep Methods:** CNNs [14-15], and RNNs [16].
 - ③ **Transformer Based Large Models:** Informer [17], AutoFormer [18], etc.
- **Strength:** Captures long-range and nonlinear dependencies.
- **Limitation:** Ignores context from event-related or semantic information.

Related Work - Textual Forecasting

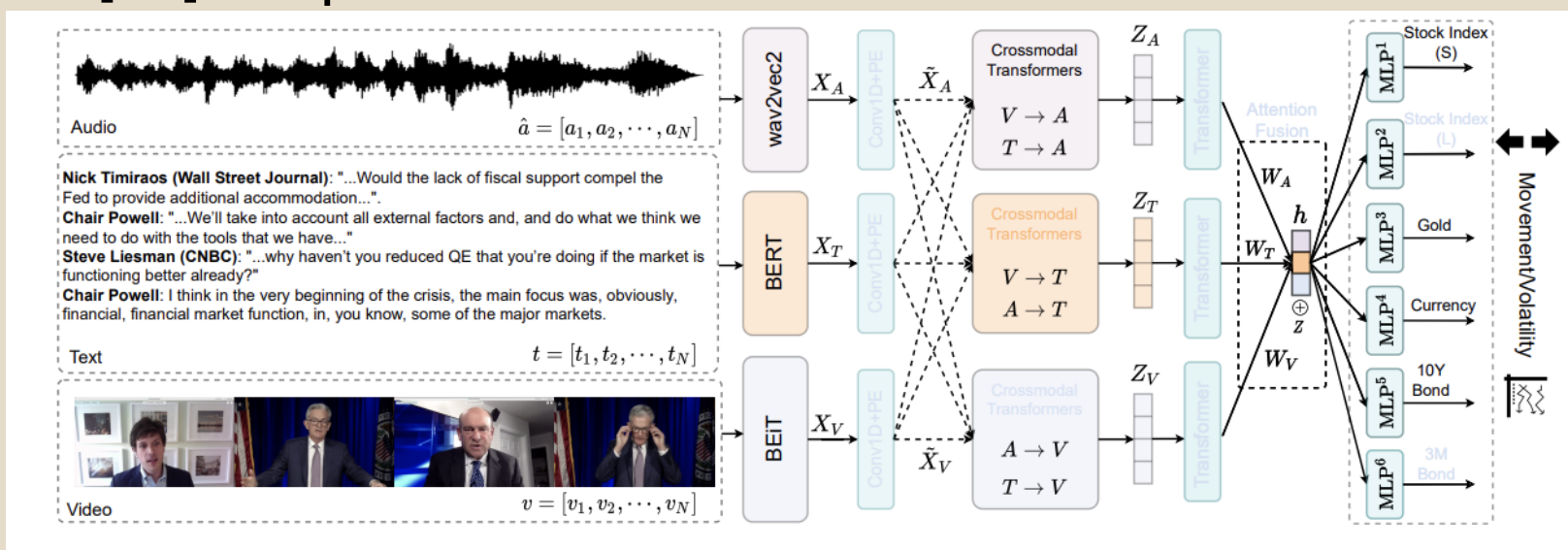
- Predicting asset price movement (binary classification) based on event-related texts.
 - ① Early methods: TF-IDF [19], Topic Models [20-21]
 - ② Deep models: RNNs [22-23]
 - ③ Pre-trained LMs: FinBERT, RoBERTa [24-25]
- **Strength:** Good at understanding event semantics.
- **Limitation:** Often lacks integration with historical price patterns; and most works merely use FOMC meeting data.



Related Work - Multi-Modal Forecasting

➤ Combine time series data with event-related textual information for more comprehensive financial forecasting..

- MONOPOLY [28] Leveraged multiple deep models, RNN for time-series, CNN for FOMC video, BERT for FOMC transcript;
- NAMEGER [29] adopted HuBERT and BEIT for textual and time-series learning.



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The multi-model neural architecture of MONOPOLY [28]

Related Work - Challenges

Challenge 1 Data Limitation: Most work only uses FOMC + daily data

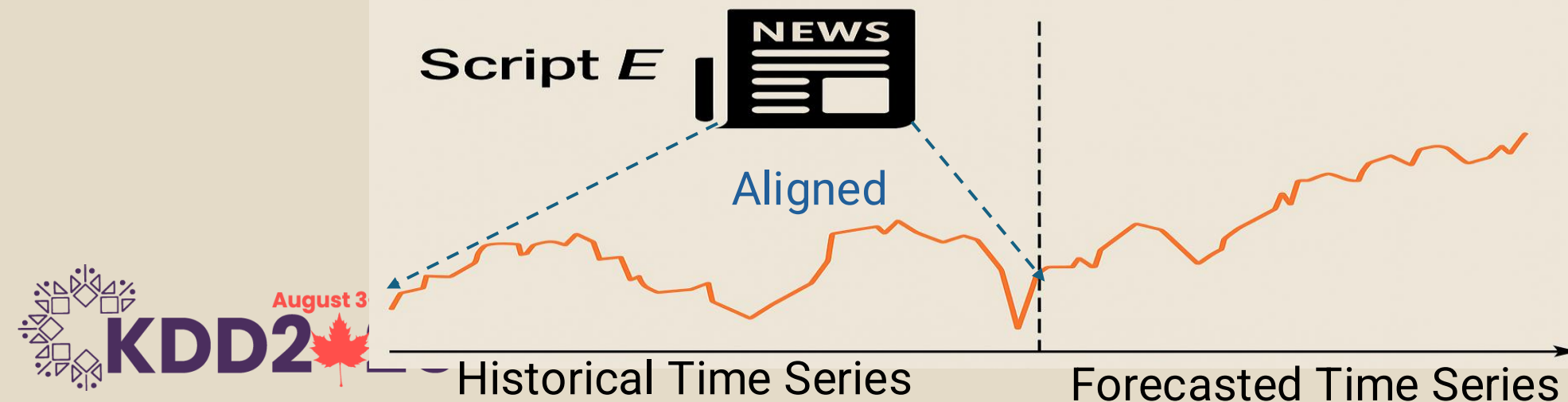
Challenge 2 Modality Limitation: Text or time-series for most works

Challenge 3 Causality Limitation: Most models lack explicit causal modeling

Problem Definition

- Forecast market trends in response to economic events by leveraging both textual policy descriptions and historical price data.

Given a dataset $U = \{[X_{i-\tau:i+\tau} \mapsto E_i]\}_{i=1}^n$ consisting of n aligned event and time-series pairs, for each data pair in U , the model uses both the event text E_i and the historical time-series segment $X_{i-\tau:i}$, which spans τ steps before the event's release at time i , to forecast the future time steps $X_{i+1:i+\tau}$.



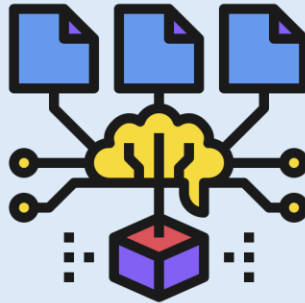
Methodology

Innovation 1: Novel Dataset



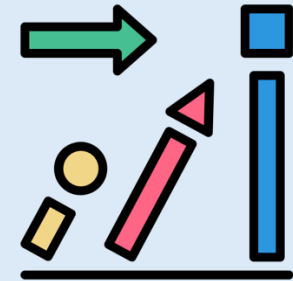
- Introduce diverse event types beyond FOMC.
- Collect 5-minute-level market data for higher temporal resolution.

Innovation 2: Multi-Model Framework



- Propose joint modeling of event texts and time series.
- Design a custom fusion and decoding mechanism.
- Apply causal learning to capture event-driven market effects.

Innovation 3: Causal Learning



- Generate counterfactual events using LLMs.
- Control sentiment levels in synthetic event scripts.
- Train model to separate causality from correlation.

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Methodology: Novel Dataset

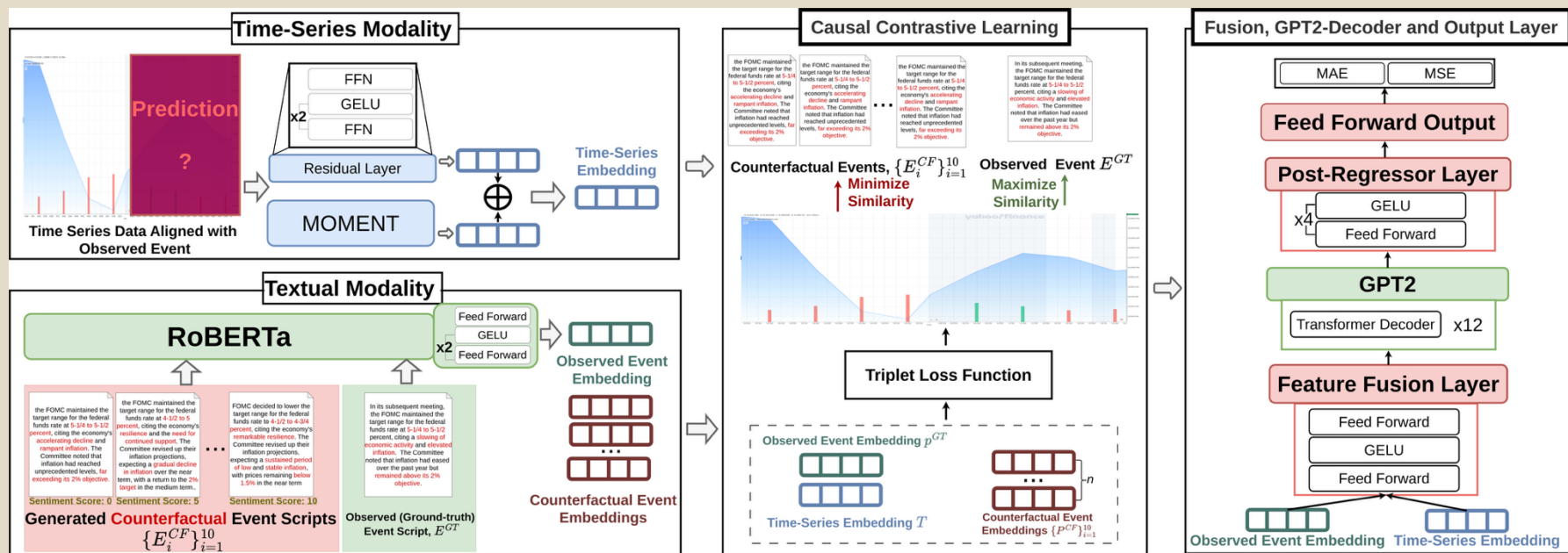
- We construct a new dataset for multimodal financial forecasting by crawling macroeconomic events and aligning them with high-frequency market data.
- **Textual Modality:** 6 key types of macro-economic events, exploited from the financial literature [30-35]
- **Time-Series Modality:** Five-minute-level data was gathered from major U.S. financial indices and bonds,

Event Type	Data Type	Frequency	Period	No. of Events	No. of C.F.s	Source
FOMC	Html	Quarterly	1993.3 ~2024.6	255	2,550	www.federalreserve.gov
Unemployment Insurance Claims	PDF, Txt	Weekly	2002.10 ~2024.6	913	9,130	oui.doleta.gov
Employment Situation	Html, Txt	Monthly	1994.2 ~2024.6	363	3,630	www.bls.gov
GDP Advance Report	Html	Monthly	1996.8 ~2024.6	333	3,330	www.bea.gov
CPI Report	Html, Txt	Monthly	1994.2 ~2024.6	357	3,570	www.bls.gov
PPI Report	Html, Txt	Monthly	1994.2 ~2024.6	348	3,480	www.bls.gov

Time Series Data	Data Types	Frequency	Range	No. of Data Points
SP500 (SPX)	Open, Close, High, Low	5 Min	2008.01 ~2024.06	331,257
Dow Industrial (INDU)	Open, Close, High, Low	5 Min	2012.07 ~2024.06	263,445
NASDAQ (NDX)	Open, Close, High, Low	5 Min	2008.01 ~2024.06	332,616
US Treasury Bond at 1-Month (USGG1M)	Open, Close, High, Low	5 Min	2013.01 ~2024.06	751,443
US Treasury Bond at 5-Year (USGG5YR)	Open, Close, High, Low	5 Min	2013.01 ~2024.06	734,773

Methodology: Framework

- We propose **CAMEF**: a **C**ausal-**A**ugmented, **M**ultimodal Framework for **E**vent-Driven Financial Forecasting.
- CAMEF integrates RoBERTa (text) and MOMENT (time series) encoders, aligns observed and counterfactual events with causal contrastive learning, and uses a GPT2 decoder to predict future market trends.



Key Features

- Joint modeling of event text and time series.
- Counterfactual generation for macro-economic events
- Contrastive learning for causal alignment
- Collection of high-frequency time series forecasting with six types of macroeconomic events

Methodology: Counterfactual Generation

- Generate counterfactual event scripts by guiding LLMs to adjust sentiment-relevant content while preserving neutral information:
- $$E'_i = E_i^{\text{neutral}} \cup f_{\text{LLM}}(E_i^{\text{sentiment}} \mid S'_i)$$

fomcminutes2...

Minutes of the Federal Open Market Committee
December 12–13, 2023

A joint meeting of the Federal Open Market Committee and the Board of Governors of the Federal Reserve System was held in the offices of the Board of Governors on Tuesday, December 12, 2023, at 10:30 a.m. and continued on Wednesday, December 13, 2023, at 9:00 a.m.¹

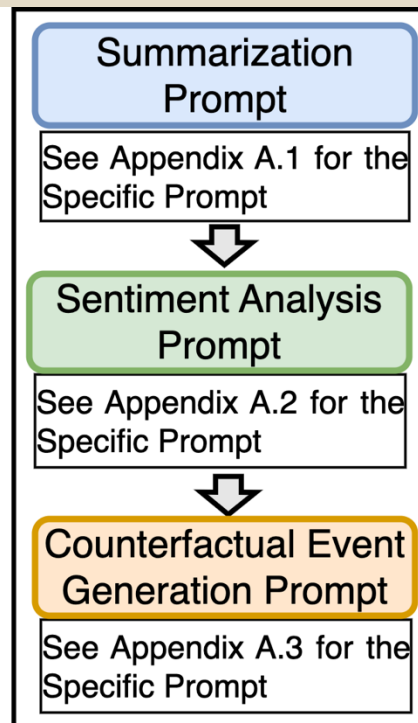
Attendance

Jerome H. Powell, Chair	Julie Ann Remache, Deputy Manager, System Open Market Account
John C. Williams, Vice Chair	Jose Acosta, Senior System Administrator II, Division of Information Technology, Board
Michael S. Barr	David Altig, Executive Vice President, Federal Reserve Bank of Atlanta
Lisa D. Cook	Alyssa Arute, ² Manager, Division of Reserve Bank Operations and Payment Systems, Board
Austan D. Goolsbee	Kimberly N. Bayard, Section Chief, Division of Research and Statistics, Board
Patrick Harker	Pendelope A. Beattie, ³ Section Chief, Office of the Secretary, Board
Philip N. Jefferson	Paola Boel, Vice President, Federal Reserve Bank of Cleveland
Neel Kashyari	David Bowman, Senior Associate Director, Division of Monetary Affairs, Board
Adriana D. Kugler	Celso Brunetti, Assistant Director, Division of Research and Statistics, Board
Lorrie K. Logan	Jennifer J. Burns, Deputy Director, Division of Supervision and Regulation, Board
Christopher J. Waller	Juan C. Clement, Special Adviser to the Board, Division of Board Members, Board
Thomas I. Barkin, Raphael W. Bostic, Mary C. Daly, and Loretta J. Mester, Alternate Members of the Committee	Edmund S. Crawley, Senior Economist, Division of Monetary Affairs, Board
Susan M. Collins and Jeffrey R. Schmid, Presidents of the Federal Reserve Banks of Boston and Kansas City, respectively	Stephanie E. Carcuro, Deputy Director, Division of International Finance, Board
Kathleen O'Neill Puse, Interim President of the Federal Reserve Bank of St. Louis	Ryan Decker, Special Adviser to the Board, Division of Board Members, Board
Joshua Gullin, Secretary	Riccardo DiCecio, Economic Policy Advisor, Federal Reserve Bank of St. Louis
Matthew M. Lucack, Deputy Secretary	Cynthia L. Doniger, Principal Economist, Division of Monetary Affairs, Board
Brian J. Bonis, Assistant Secretary	Rochelle M. Edge, Deputy Director, Division of Monetary Affairs, Board
Michelle A. Smith, Assistant Secretary	
Mark E. Van Der Weide, General Counsel	
Richard Ostrander, Deputy General Counsel	
Trevor A. Reeve, Economist	
Stacey Tevlin, Economist	
Beth Anne Wilson, Economist	
Shaghi Ahmed, Roc Armenter, James A. Clouse, Eric M. Engen, Anna Paulson, Andrea Raffio, Chiara Scotti, and William Wascher, Associate Economists	
Roberto Perli, Manager, System Open Market Account	

¹ The Federal Open Market Committee is referenced as the "FOMC" and the "Committee" in these minutes; the Board of Governors of the Federal Reserve System is referenced as the "Board" in these minutes.

² Attended through the discussion of developments in financial markets and open market operations.

³ Attended through the discussion of the economic and financial situation.



... In its subsequent meeting, the FOMC maintained the target range for the federal funds rate at 5-1/4 to 5-1/2 percent, citing a **slowing of economic activity** and **elevated inflation**. The Committee noted that inflation had eased over the past year but **remained above its 2% objective**. ...

Observed Event Script, E^{GT}



Sentiment rating: 4, Explanation: The sentiment of this FOMC summary is slightly negative. The language used is cautious and neutral, with a focus on the Committee's efforts to balance the need to support economic growth with the need to address elevated inflation pressures....

Sentiment Analysis Output



the FOMC maintained the target range for the federal funds rate at 5-1/4 to 5-1/2 percent, citing the economy's **accelerating decline** and **rampant inflation**. The Committee noted that inflation had reached unprecedented levels, **far exceeding its 2% objective**.

Counterfactual Event
@ Sentiment Score: 0

...

FOMC decided to lower the target range for the federal funds rate to 4-1/2 to 4-3/4 percent, citing the economy's **remarkable resilience**. The Committee revised up their inflation projections, expecting a **sustained period of low and stable inflation**, with prices remaining **below 1.5%** in the near term

Counterfactual Event
@ Sentiment Score: 10

Original FOMC Meeting Report

Prompt Pipeline via LLaMa3

Generated Output from LLM



Methodology: Learning Objectives

➤ Forecasting Losses:

Input: E_i (event script), T_i (time series)

CAMEF: $\hat{y}_i = \text{CAMEF}(E_i, T_i)$

Losses: $\mathcal{L}_{\text{MSE}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$, $\mathcal{L}_{\text{MAE}} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$

➤ Contrastive Learning for Ground-Truth and Counterfactual Events:

$$\mathcal{L}_{\text{Causal-CL}} = \max(0, d(\mathbf{E}_i^{\text{GT}}, \mathbf{T}) - d(\mathbf{E}_i^{\text{CF}}, \mathbf{T}) + \alpha)$$

➤ Combined Learning Objective

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Forecast}} + \mathcal{L}_{\text{Causal-CL}}$$

Experiment

➤ **We evaluate CAMEF through three key research questions:**

□ **RQ1 Accuracy:**

How well does CAMEF forecast the market based on event-driven inputs?

□ **RQ2 Model Effectiveness:**

How do different components contribute to prediction performance?

□ **RQ3 Event Analysis:**

Which event types have stronger influence on market movements?

Experiment

➤ RQ1 Accuracy:

- **Baselines (Uni-Model and Multi-Model):** ARIMA, DLinear, Autoformer, FEDformer, iTransformer, PatchTST, GPT4MTS, etc.
- **Metrics:** MSE / MAE
- **Forecast horizons:** 35 (0.5 day), 70 (1 day), 140 (2 days) steps
- **CAMEF consistently outperforms all baselines**

Model / Datasets	Forecasting Length	SP500 (SPX)		Dow Industrial (INDU)		NASDAQ (NDX)		USGG1M		USGG5YR	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ARIMA	35	0.0032628	0.0308016	0.0121513	0.0523810	<u>0.0065253</u>	0.0435694	0.0072223	0.0262512	0.0038973	0.0340227
	70	0.0035361	0.0352004	0.0139245	0.0656002	<u>0.0082324</u>	0.0520933	0.0028710	0.0304132	0.0039441	0.0366630
	140	0.0051080	0.0439935	0.0219147	0.0793471	<u>0.0118692</u>	0.0665155	0.0089949	0.0338275	0.0050746	0.0455617
DLinear	35	0.0144331	0.0896910	0.0395136	0.1380539	0.0183989	0.0999178	0.0108576	0.0706170	0.0147211	0.0876379
	70	0.0120578	0.0817373	0.0406573	0.1282699	0.0189431	0.0972439	0.0093591	0.0678380	0.0146430	0.0881574
	140	0.0178138	0.0931039	0.0747210	0.1724027	0.0344153	0.1287803	0.0101465	0.0645396	0.0179185	0.0977673
Autoformer	35	0.0068136	0.0540556	0.0277636	0.0975948	0.0135249	0.0796486	0.0047505	0.0388076	0.0092717	0.0622862
	70	0.0088997	0.0628341	0.0375279	0.1185710	0.0185264	0.0933398	0.0065554	0.0471531	0.0113750	0.0727373
	140	0.0158248	0.0829188	0.0772580	0.1640365	0.0334504	0.1262163	0.0086212	0.0533337	0.0152298	0.0875821
FEDformer	35	0.0072377	0.0576221	0.0304808	0.1044447	0.0128668	0.0758742	0.0063596	0.0519141	0.0094995	0.0494306
	70	0.0088056	0.0621386	0.0399824	0.1229772	0.0171008	0.0889995	0.0062841	0.0448822	0.0099615	0.0664197
	140	0.0157429	0.0819469	0.0786426	0.1677705	0.0313433	0.1214097	0.0083784	0.0518365	0.0131231	0.0782861
iTransformer	35	0.0064209	0.0516341	0.0270860	0.0927605	0.0125008	0.0751656	0.0011000	0.0155975	0.0056660	0.0183524
	70	0.0069612	0.0540920	0.0304566	0.1038424	0.0151214	0.0816262	0.0021811	0.0221315	0.0011721	0.0226975
	140	0.0128021	0.0718771	0.0680991	0.1486782	0.0254479	0.1047119	0.0052255	0.0327429	0.0017441	0.0282537
PatchTST	35	0.0063304	0.0507462	0.0293131	0.0989455	0.0122679	0.0764552	0.0012060	0.0163610	0.0063078	0.0520734
	70	0.0072471	0.0547738	0.0339444	0.1116023	0.0153753	0.0824677	0.0021643	0.0223036	0.0079617	0.0606007
	140	0.0130219	0.0712171	0.0688582	0.1452592	0.0256001	0.1047749	0.0054544	0.0341517	0.0118553	0.0747537
GPT4MTS	35	0.00795088	0.0674255	<u>0.0026558</u>	0.0417553	0.0011035	0.0240469	0.0017016	0.0309320	0.0028713	0.0389277
	70	0.00171038	0.0305950	<u>0.0027033</u>	<u>0.0393112</u>	0.0016205	0.0336116	0.0019098	0.0279222	0.0023371	0.0354777
	140	0.00212612	<u>0.0330497</u>	<u>0.0045029</u>	<u>0.0458267</u>	0.0025175	0.0341671	0.0013648	0.0260887	0.0037004	0.0449272
TEST	35	<u>0.00073333</u>	<u>0.0199733</u>	0.0026572	0.0296887	0.0006593	0.0194091	0.0003252	0.0137511	0.0013633	0.0265036
	70	<u>0.00078762</u>	<u>0.0205412</u>	0.0088903	0.0599179	0.0010678	<u>0.0248594</u>	<u>0.0010515</u>	<u>0.0182706</u>	0.0034630	0.0415002
	140	<u>0.00278572</u>	0.0467150	0.0070749	0.0557744	0.0020130	<u>0.0362325</u>	0.0006995	<u>0.0207850</u>	0.0024356	0.0375164
CAMEF	35	0.00048860	0.0154050	0.0025349	<u>0.0366245</u>	0.0005468	0.0178845	0.0002883	0.0118010	0.0013234	0.0260618
	70	0.00064780	0.0178691	0.0025042	0.0365500	0.0005814	0.0162882	0.0004402	0.0139699	<u>0.0020701</u>	<u>0.0326371</u>
	140	0.0010756	0.0210284	0.0039313	0.0383459	0.0010159	0.0207716	0.0004938	0.0148485	<u>0.0022458</u>	<u>0.0336680</u>

Forecasting Test of CAEMF against SOTA baselines across 5 datasets

Experiment

➤ RQ2: Model Effectiveness:

- We conduct ablation studies to evaluate the contribution of each CAMEF component:

Textual	Causal	Feature Fusion	GPT2 Decoder	Post-Regressor	SPX			INDU			NDX			USGG1M			USGG5YR			AVERAGE		
					35	70	140	35	70	140	35	70	140	35	70	140	35	70	140	35	70	140
✗	✓	✓	✓	✓	0.00074	0.00340	0.00120	0.00340	0.00330	0.01033	0.00131	0.00097	0.00197	0.00080	0.00179	0.00092	0.00222	0.00276	0.00380	0.00169	0.00199	0.00364
✓	✗	✓	✓	✓	0.00068	0.00095	0.00106	0.02939	0.00281	0.00533	0.00064	0.00073	0.00121	0.00065	0.00083	0.00099	0.00160	0.00220	0.00308	0.00659	0.00170	0.00233
✓	✓	✗	✓	✓	0.00080	0.00079	0.00110	0.01173	0.00391	0.00581	0.00069	0.00073	0.00168	0.00047	0.00046	0.00216	0.00251	0.00306	0.00426	0.00324	0.00212	0.00300
✓	✓	✓	✗	✓	0.00073	0.00067	0.00114	0.26768	0.72387	0.18091	0.00062	0.00069	0.00158	0.00043	0.02110	0.75057	0.00220	0.00309	0.00566	0.05433	0.14593	0.03801
✓	✓	✓	✓	✗	0.00662	0.03911	0.00979	0.50841	0.57007	0.22267	0.00774	0.00852	0.00950	0.28932	0.02110	0.75057	0.56705	0.06188	0.08251	0.27583	0.14014	0.21501
Full CAMEF Model					0.00048	0.00064	0.00107	0.00253	0.00250	0.00393	0.00054	0.00058	0.00101	0.00028	0.00044	0.00049	0.00132	0.00207	0.00224	0.00104	0.00124	0.00174

- Each module contributes to performance. The full CAMEF model consistently achieves the lowest average MSE across all markets and horizons.

Experiment

➤ RQ3 Event Analysis:

- How different types of economic events influence the forecasting accuracy.
- CAMEF is trained with isolated event types to assess their individual predictive strength.
- **FOMC Minutes** and **Employment Situation Reports** are most informative

Event Type	Forecasting Length = 35		Forecasting Length = 70		Forecasting Length = 140	
	MSE	MAE	MSE	MAE	MSE	MAE
Unemployment Insurance	0.0004870 ↓	0.0159497 ↑	0.0005199 ↓	0.0157778 ↓	0.0006948 ↓	0.0190141 ↓
Employment Situation	0.0003923 ↓	0.0142684 ↓	0.0004612 ↓	0.0154431 ↓	0.0013767 ↑	0.0218677 ↑
GDP Advance	0.0006203 ↑	0.0175397 ↑	0.0005897 ↓	0.0175464 ↓	0.0011554 ↑	0.0217548 ↑
FOMC Minutes	0.0003401 ↓	0.0127694 ↓	0.0004448 ↓	0.0170094 ↓	0.0006433 ↓	0.0185657 ↓
CPI Report	0.0005645 ↑	0.0160723 ↑	0.0010660 ↑	0.0212536 ↑	0.0008187 ↓	0.0197824 ↓
PPI Report	0.0005275 ↑	0.0148434 ↓	0.0008054 ↑	0.0201844 ↑	0.0017646 ↑	0.0251856 ↑
Full Selection	0.0004886	0.0152405	0.0006478	0.0178691	0.0010756	0.0210284

- In contrast, **GDP Advance** and **PPI Reports** yield weaker predictive performance, particularly over longer horizons (140 steps).

Conclusion

- **CAMEF** introduces a **causal-augmented, multimodal** forecasting framework that effectively integrates policy texts and market time series.
- CAEMF outperforms SOTA unimodal and multimodal large models in predictive accuracy.
- Extensive evaluations demonstrate the architecture robustness.
- FOMC is found to be the most informative event in financial market.

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